BeMaMaJa

Group Project Report – INFO4271 Modern Search Engines

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Code repository: https://github.com/BeMaMaJa-Search-Engine/BeMaMaJa-Search-Engine.git

Abstract

1 Data Collection and Crawling

1.1 Corpus scope and seeds

1.2 Crawler design

1.3 Filtering and stored page representation

Table 1: Final corpus summary.

Saved English pages	Attempted URLs
Pages with body text	Unique domains
Median body length	Crawl duration

2 Preprocessing and Index Construction

2.1 Shared preprocessing contract

2.2 Manual inverted index

2.3 Build process and validation

Table 2: Index statistics and build characteristics.

Measure	Value
Indexed documents	
Vocabulary size	
Average body length	
Documents with outgoing links	
Index size / build time	

2.4 Engineering justification

3 Retrieval and Re-ranking

3.1 Manual BM25 first stage

$$\mathrm{BM25}(d,q) = \sum_{t \in q} \log\left(1 + \frac{N - df_t + 0.5}{df_t + 0.5}\right) \frac{tf_{t,d}(k_1 + 1)}{tf_{t,d} + k_1\left(1 - b + b\frac{|d|}{\mathrm{avgdl}}\right)}.$$

3.2 Query processing

3.3 Second-stage innovation

3.4 Diagnostic evaluation

Table 3: Ablation or diagnostic comparison.

Configuration	Query 1	Query 2	Interpretation
BM25 only			
BM25 + innovation			

4 Interface and Evaluation

Figure 1.

4.1 Interactive interface

4.2 Batch and effectiveness evaluation

Table 4: Final evaluation checklist.

Requirement	Evidence
Five queries, 100 URLs each	
Average / maximum runtime	
Final mean nDCG	
Repository branch frozen	

4.3 Limitations and reproducibility